

CHAPTER 8

Software Maintenance

Outline

- **Concept of software maintenance**
- **Types of software maintenance**
- **Costs of software maintenance**
- **Maintainability**
- **Process of maintenance**

What is software maintenance ?

- ✓ **Modifying a program after it has been put into use**
- ✓ **Modifications or corrections made to a software system after it has been released to its customers**
- ✓ **Changing a software system while it is in operation**
- ✓ **Evolving a software system to adapt to changing business conditions and user needs**
- ✓ **Managing the processes of system change**

软件维护的概述

- 软件维护是软件开发工作完成以后，在用户使用期间，对软件所做的补充、修改和增加工作。
- 软件运行=软件维护
- 在软件维护中，为增加和改进软件的功能所做的维护占80%，而为改正错误所做的维护仅占20%。
- 统计数据表明：实际上用于软件维护的费用占软件总费用的55-80%。
- 软件维护比软件开发更困难，需要更多的创造性工作。
- 一般不涉及体系结构的重大变化

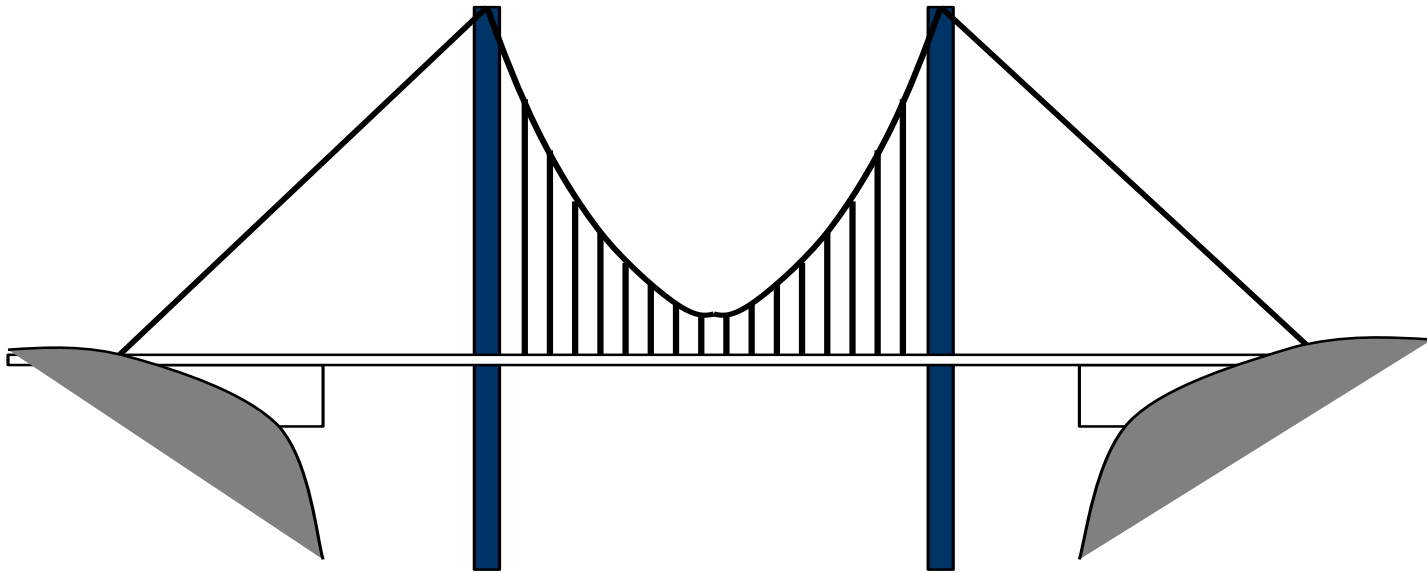
Why software maintenance ?

- 从机械、土木等工程中的启发
- 软件系统中的变化
- 软件维护是不可避免的

Why software maintenance ?

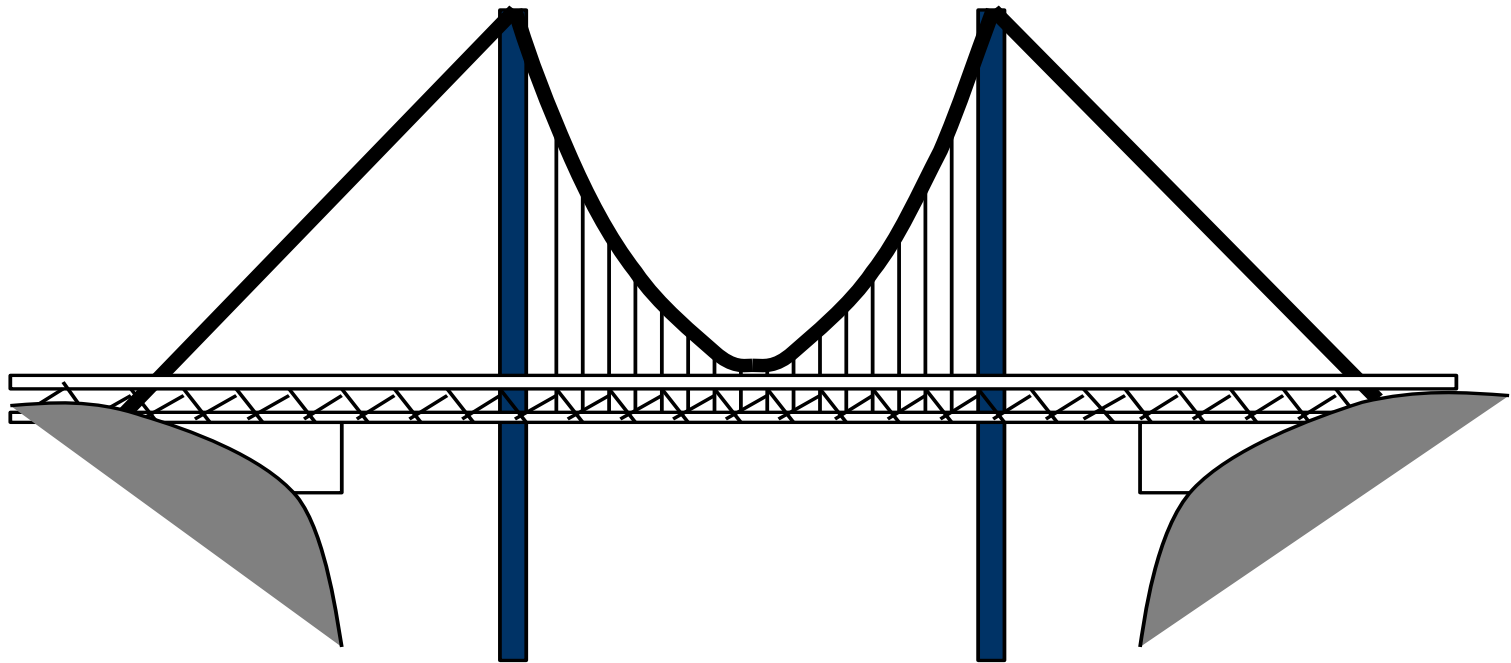
从机械、土木等工程中的启发

The Beginning Product

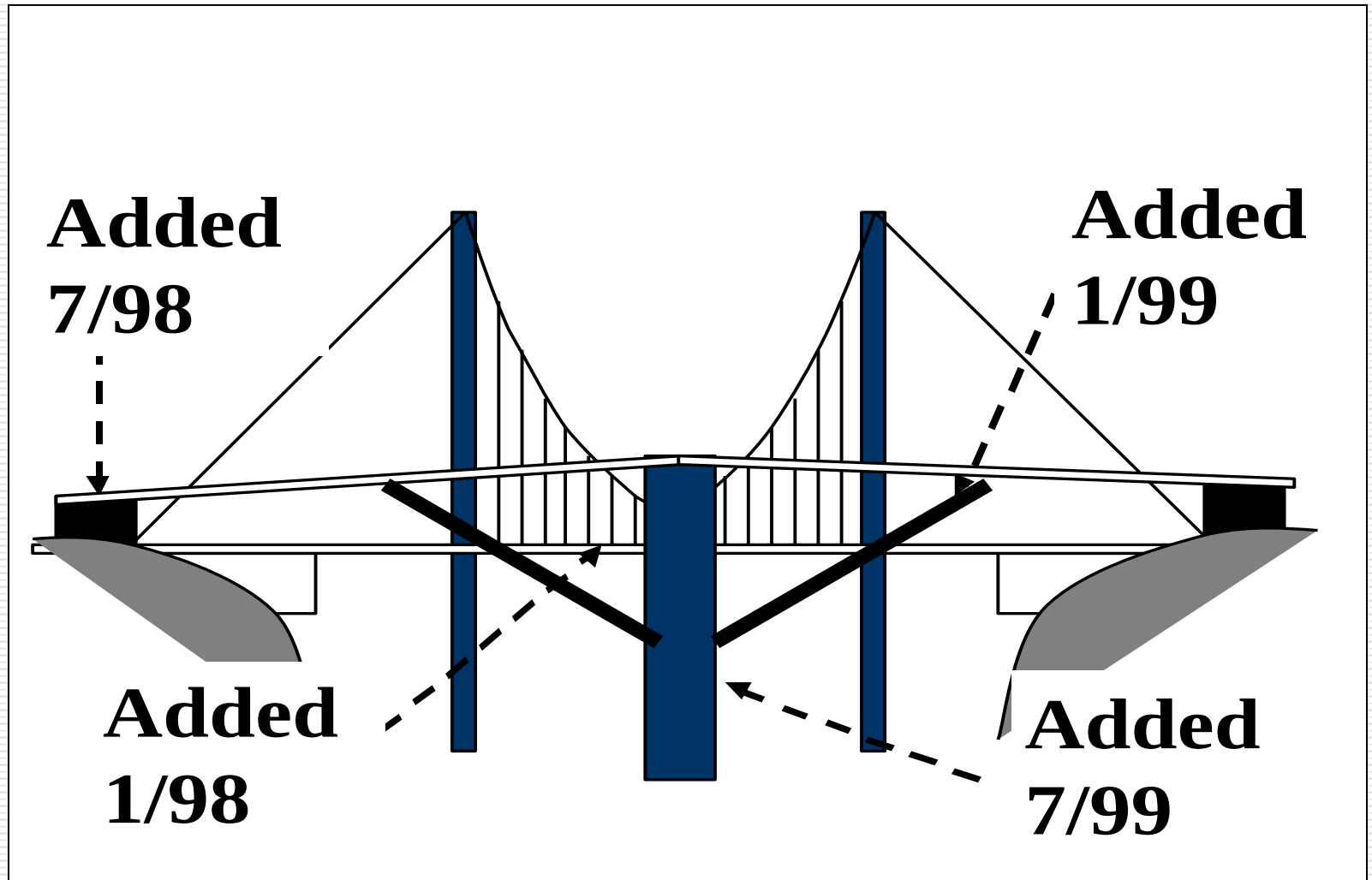


Why software maintenance ?

Expansion



Why software maintenance ?



The First Law of Software Engineering

“No matter where you are in the system life cycle, the system will **change, and the desire to change it will persist throughout the life cycle”**

Bersoff et al. (1980)

软件系统，经常发生：变化，变更，改变，转变，...

“变”

“变” 的根源

- ✓ New **business** or market conditions which cause changes in product requirements or business rules
- ✓ New **customer** needs that demand modification of data, functionality or services delivered by the system
- ✓ **Reorganisation** and/or business downsizing that changes priorities and **team** structures
- ✓ **Budgetary** or **scheduling** constraints that cause a redefinition of the system

Software change

- ✓ **New requirements emerge when the software is used**
- ✓ **Errors must be repaired**
- ✓ **New equipment must be accommodated**
- ✓ **The performance or reliability may have to be improved**
- ✓ **MOST CHANGES ARE JUSTIFIED**
- ✓ **A key problem for organisations is implementing and managing change to their legacy systems**

软件变化原因

软件的变化是不可避免的

- ✓ 软件在使用过程中，新的需求不断出现
- ✓ 商业环境在不断地变化
- ✓ 业务逻辑在不断地变化
- ✓ 软件中的缺陷需要进行修复
- ✓ 计算机硬件和软件环境的升级需要更新现有的系统
- ✓ 软件的性能和可靠性需要进一步改进

关键：

采取适当的策略，有效地实施和管理软件的变化！

Maintenance is inevitable

- **The system requirements are likely to change while the system is being developed because the environment is changing. Therefore a delivered system won't meet its requirements!**
- **Systems are tightly coupled with their environment. When a system is installed in an environment it changes that environment and therefore changes the system requirements.**
- **Systems MUST be maintained therefore if they are to remain useful in an environment.**

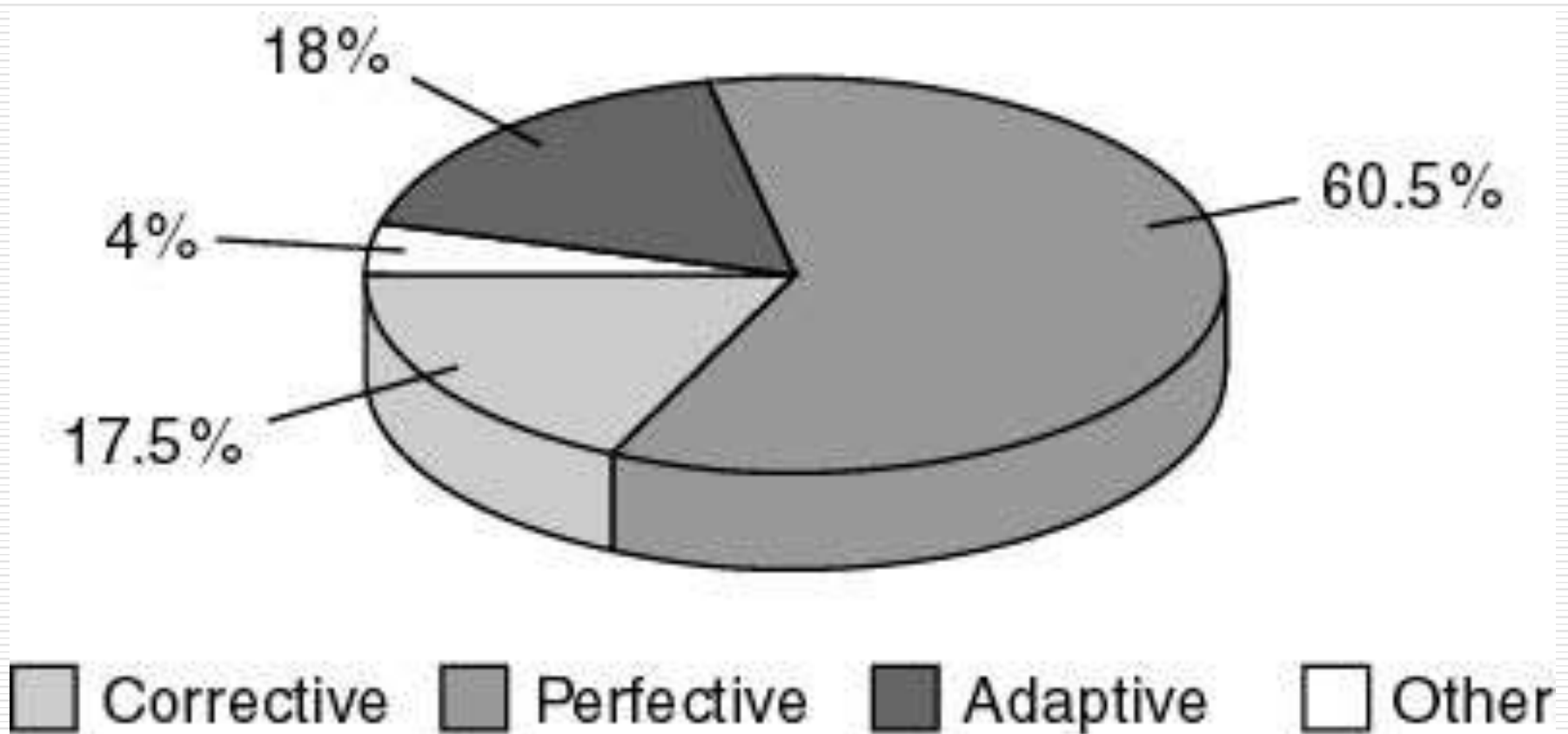
Types of Maintenance

- **Commonly divided into four main categories**
 - **Corrective Maintenance**
 - **Adaptive Maintenance**
 - **Perfective Maintenance**
 - **Preventative Maintenance**

维护种类

- **改正性维护：** 目的是识别和矫正功能错误、性能错误和实现上的错误。
- **适应性维护：** 使软件适应于外界环境的改变而对软件所做的修改工作。
- **完善性维护：** 为了扩充软件的功能或改善软件的性能对软件所做的改变。
- **预防性维护：** 为了以后更便于维护，或者为了改进可靠性，或者提供更好的基础便于将来提高性能而修改软件。

Distribution of maintenance effort



改正性维护

□ 什么是纠错性维护？

- 诊断和纠正软件中的错误或者缺陷

□ 起因

- 用户在使用软件过程中发现错误

□ 目的

- 改正软件系统中潜藏的错误

改正性维护

- ✓ **Focused on fixing failures**
- ✓ **Is a reactive process**
 - **failures and their associated faults generally need to be corrected either immediately or in the near future**
- ✓ **改错的成本不同**
 - **Coding - usually relatively cheap**
 - **Design - more expensive as they may require changes to several program components**
 - **Requirements - most expensive - may require extensive system redesign**
- ✓ **Requirements and Design are the source of approximately 80% of failures**

改正性维护

- ✓ Fixing a fault has a 20 to 50% chance of introducing another fault
- ✓ Reasons for new faults include
 - the ripple effect, where a change in one area may cause changes in seemingly unrelated areas
 - Person who makes the repair is generally not the person who wrote the code or designed the system
- ✓ Two types of corrective maintenance
 - **Emergency Repairs** - short time frame, often a single program, failure needs to be repaired as soon as possible
 - **Scheduled Repairs** - failure doesn't need immediate attention, re-examination of all emergency repairs

适应性维护

□ 什么是适应性维护？

- 对软件进行改造以适应新的运行环境和平台

□ 原因

- 软件运行于环境(硬件、OS、网络等)之上
- 软件服役时间较长，运行环境变化很快

□ 目的

- 适应环境变化和发展而对软件进行维护

适应性维护

- ✓ **The evolution of the system to meet the needs of the user and the business.**
- ✓ **Caused by**
 - **internal needs**
 - **external competition**
 - **external requirements e.g. changes in law**
- ✓ **Essentially we are introducing new requirements to the system.**
- ✓ **Therefore we should treat like a new development in our approach and methods.**

完善性维护

□ 什么是完善性维护？

- 对软件进行改造以增加新功能、修改已有功能

□ 原因

- 用户要求增加新功能、建议修改已有功能或提出其他改进意见

□ 目的

- 满足用户日益增长的各种需求

完善性维护

- ✓ **Old proverb says “If it is not broken, don’t fix it”**
- ✓ **Perfective maintenance ignores this ancient piece of wisdom**
- ✓ **Is about improving the quality of a program that already works**
- ✓ **Aim to achieve**
 - **reduced costs in using the system**
 - **increase maintainability of the system**
 - **more closely meet the users’ needs**

完善性维护

- ✓ Includes all efforts to polish or refine the quality of the software or the documentation
- ✓ Important that the potential benefits of the perfective maintenance outweigh
 - the costs of the maintenance
 - and the opportunity costs of improvements elsewhere or using the resources on new developments
- ✓ Before performing perfective maintenance, one should go through an analysis process
- ✓ Nevertheless, a little perfective maintenance can have dramatic effects

预防性维护

□ 什么是预防性维护？

- 对软件的结构进行改造以便提高软件的可靠性和可维护性等

□ 原因

- 为改善软件系统的可维护性和可靠性，为以后的软件改进奠定基础

□ 目的

- 获取和改善软件结构

预防性维护

- ✓ Can be seen as radical perfective maintenance or as an alternative to maintenance
- ✓ More commonly known as Software Re-engineering
- ✓ Taking a legacy system and converting its structure or converting to a new language
- ✓ Old system starts as a specification for the new system
- ✓ Common method now is known as wrappers where an entire system is placed in an OO wrapper and treated as one large object

维护成本

- ✓ Usually greater than development costs (2* to 100* depending on the application).
- ✓ Affected by both technical and non-technical factors.
- ✓ Increases as software is maintained.
Maintenance corrupts the software structure so makes further maintenance more difficult.
- ✓ Ageing software can have high support costs (e.g. old languages, compilers etc.)

The Cost of Maintenance

➤ 通常认为:

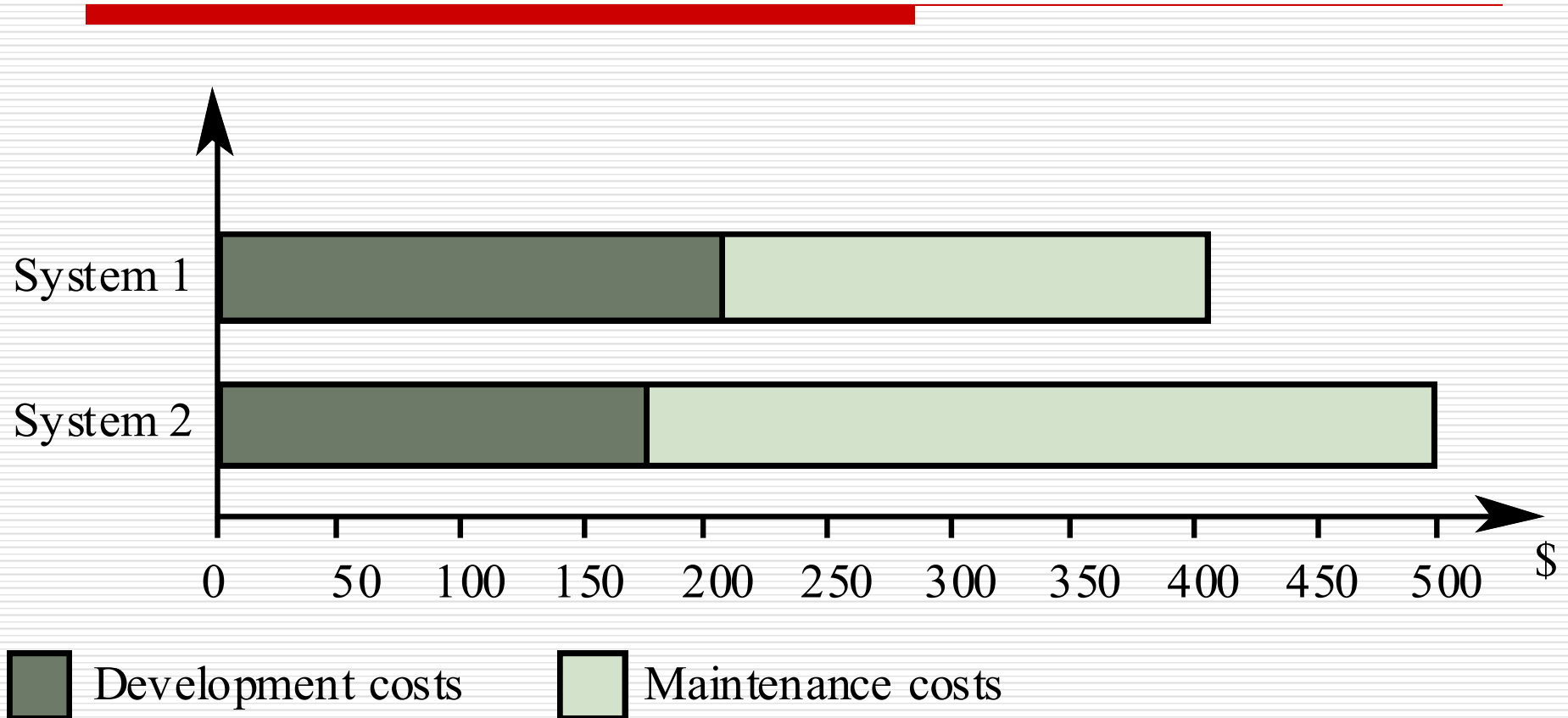
● Requirements Definition	3%	
● Preliminary Design	3%	
● Detailed Design	5%	
● Implementation	7%	■ 70年代 (35%-40%)
● Testing	15%	■ 80年代 (60%左右)
● Maintenance	67%	■ 90年代 (75%左右)
		■ 21世纪 (80%左右)

➤ Another study found at least 50% of effort spent on maintenance

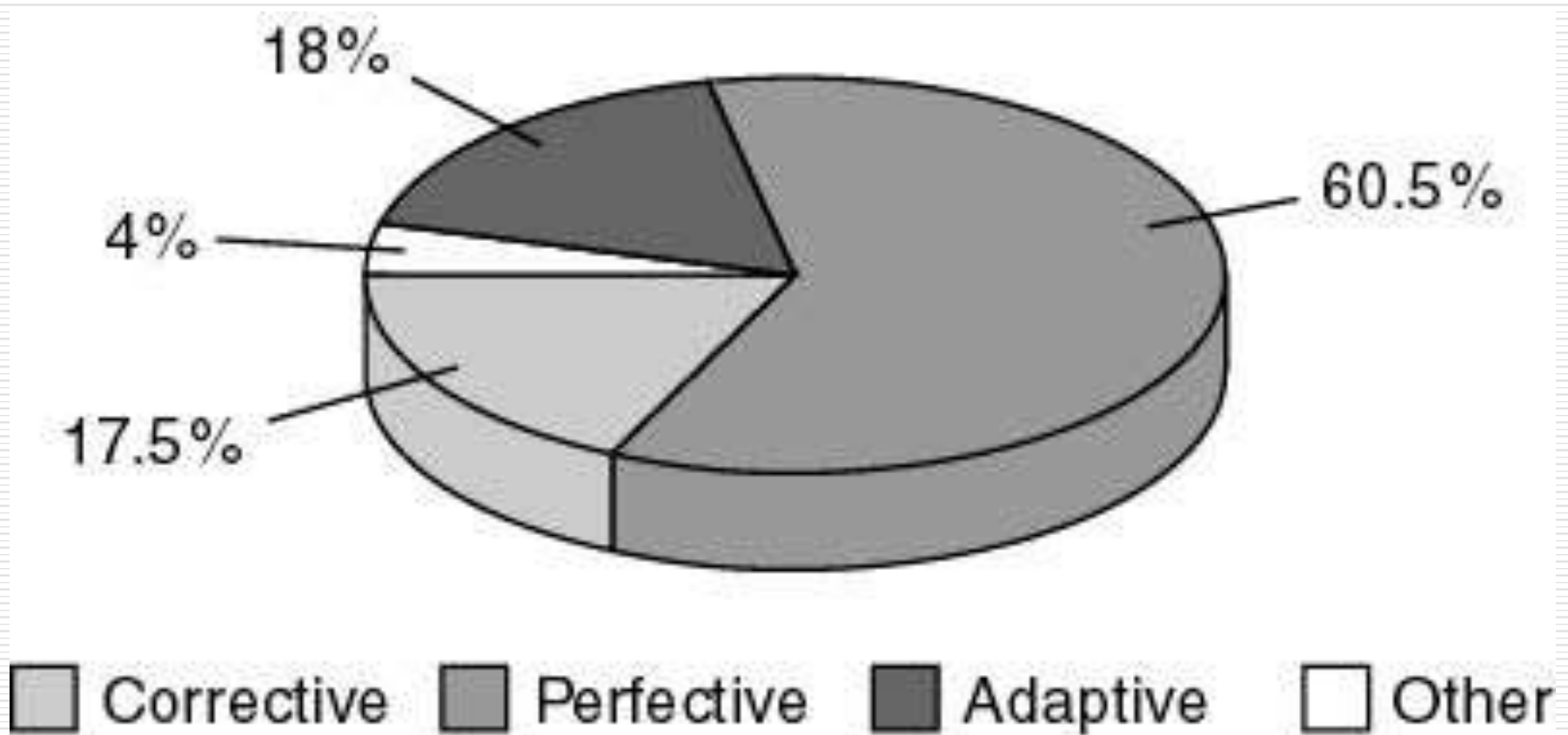
➤ Another study found between 65% and 75% on maintenance

➤ In embedded real-time systems, maintenance costs may be up to 4 times development costs

Development/maintenance costs



Distribution of maintenance effort



维护的成本

$$M = P + K e^{(c-d)}$$

M : 维护工作总工作量

P : 生产性工作量

K : 经验常数

c : 复杂度

d : 对该软件熟悉程度的度量

Why is Maintenance so Costly?

- ✓ Most software is between 10 and 15 years old
- ✓ Much of that software is showing its age as it was created when program size and storage space were far more important factors
- ✓ This has lead to inflexible designs, coding and documentation
- ✓ Maintenance is usually done by inexperienced staff unfamiliar with the application
- ✓ Developers don't like maintenance
- ✓ Changes often cause new faults in the system
- ✓ Changes tend to degrade the structure of a program
- ✓ Changes are often not documented

为什么维护工作成本很高

- (1) 成功的软件，生命周期较长，10-15年
- (2) 难以跟踪软件版本的进化过程, 软件的变化未在文档中反映出来.
- (3) 难以跟踪软件的创建过程, 无文档或不全.
- (4) 开发者不喜欢维护.
- (5) 难以读懂他人程序.
- (6) 软件人员流动性大.
- (7) 设计时未考虑修改需要, 修改困难.
- (8) 维护工作无吸引力, 缺乏成就感.
- (9) 维护可能引入新的错误!

软件可维护性

➤ 软件可维护性的定义

软件可维护性是指软件被理解、改正、调整和改进的程度

➤ 衡量软件质量的几个主要特性

- ✓ 所采取的开发方法（OS, OO）
- ✓ 开发人员素质
- ✓ 文档是否齐全
- ✓ 可维护性
- ✓ 可使用性
- ✓ 可靠性
- ✓ 标准化……

可维护性的度量

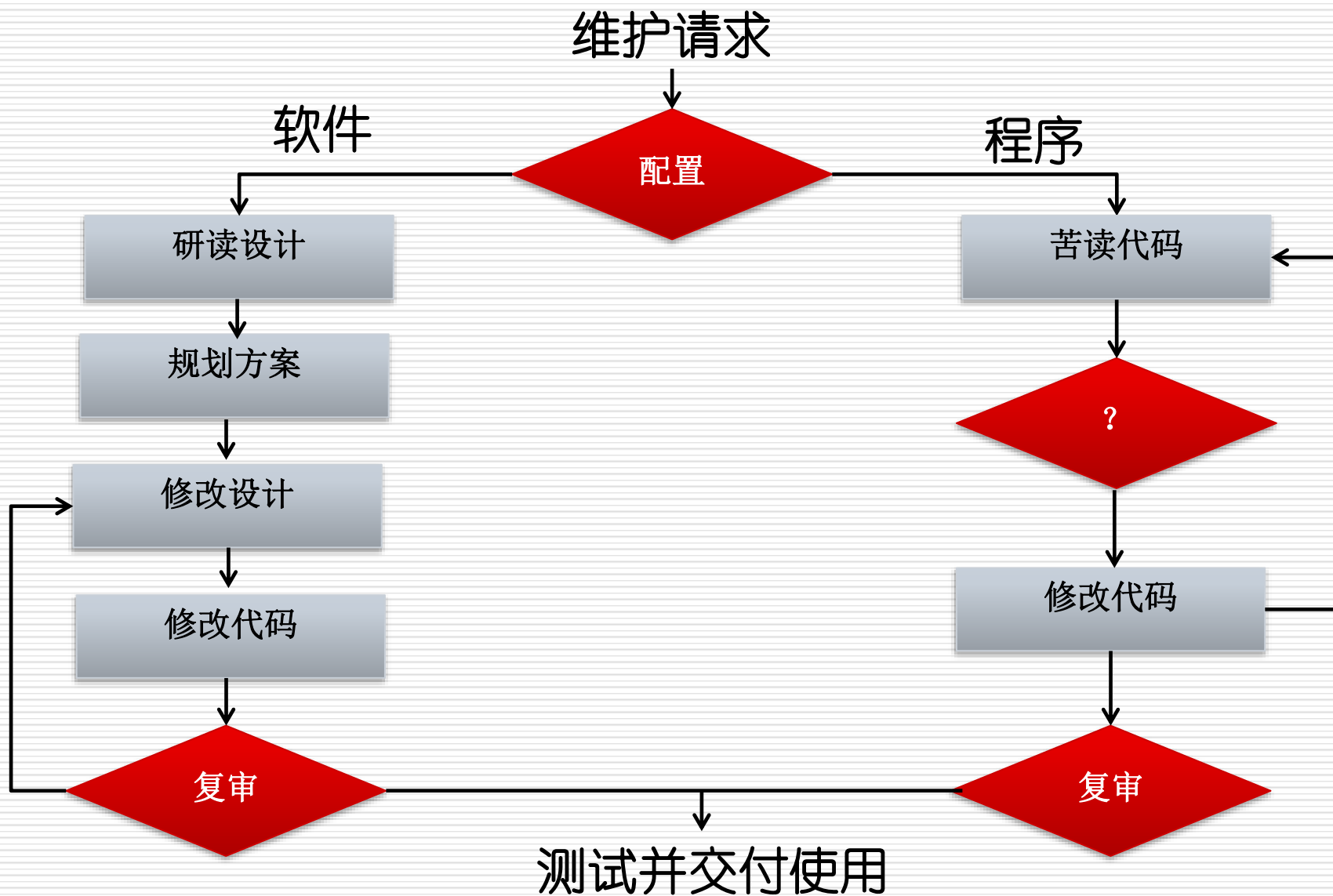
度量程序可维护性的7个特性

	改正性维护	适应性维护	完善性维护
可理解性			
可测试性			
可修改性			
可靠性			
可移植性			
可使用性			
效 率			

提高可维护性的方法

- ✓ 建立明确的软件质量目标和优先级
- ✓ 使用提高软件质量的技术和工具
- ✓ 进行明确的质量保证审查
- ✓ 选择可维护的程序设计语言
- ✓ 改进程序的文档
- ✓ 提倡规范化、标准化开发
- ✓ 开发软件时考虑到可维护要求

软件维护过程



维护过程

维护机构机构

维护申请报告

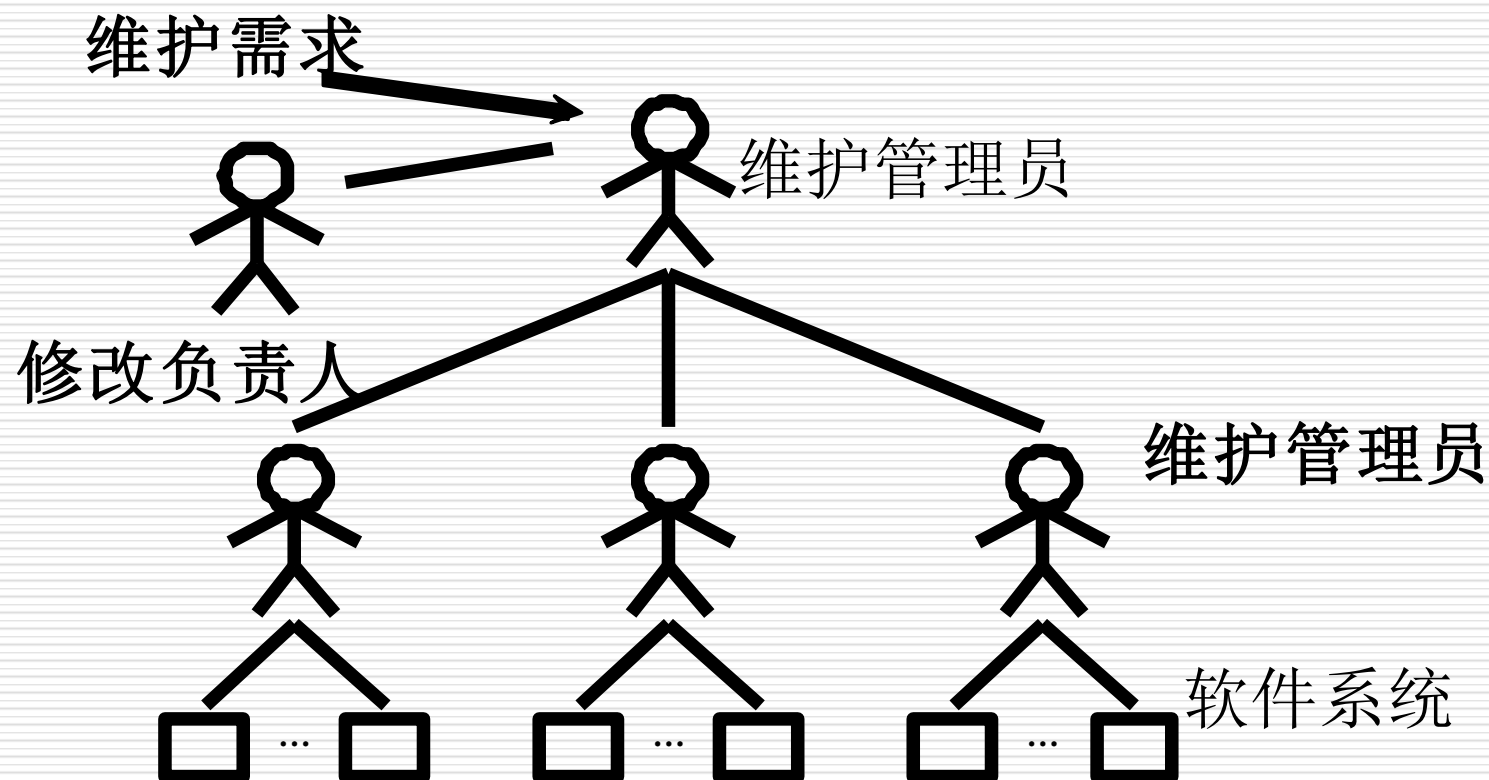
维护的工作流程

维护记录

维护评价

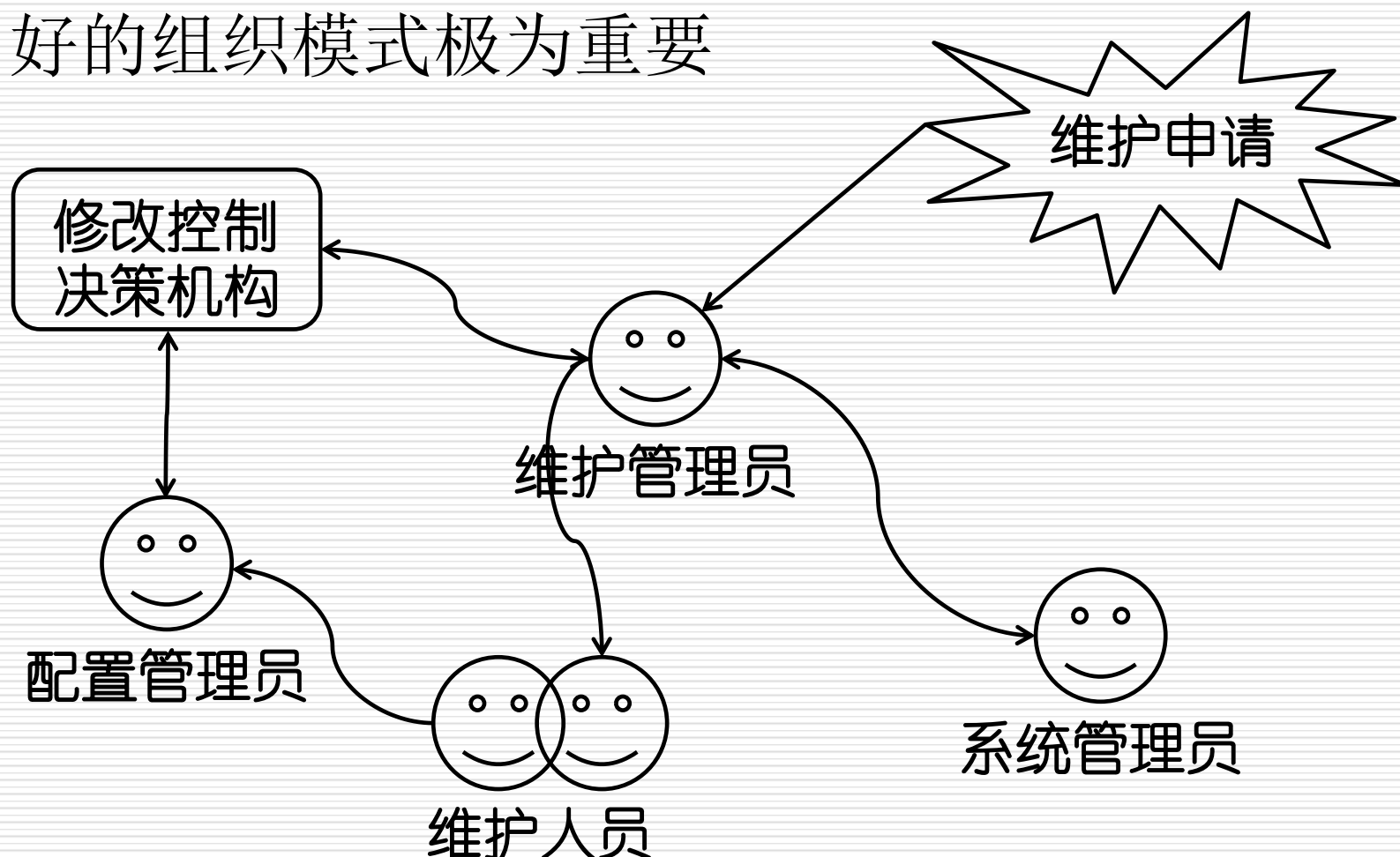
完成

维护的组织机构



成立维护组织

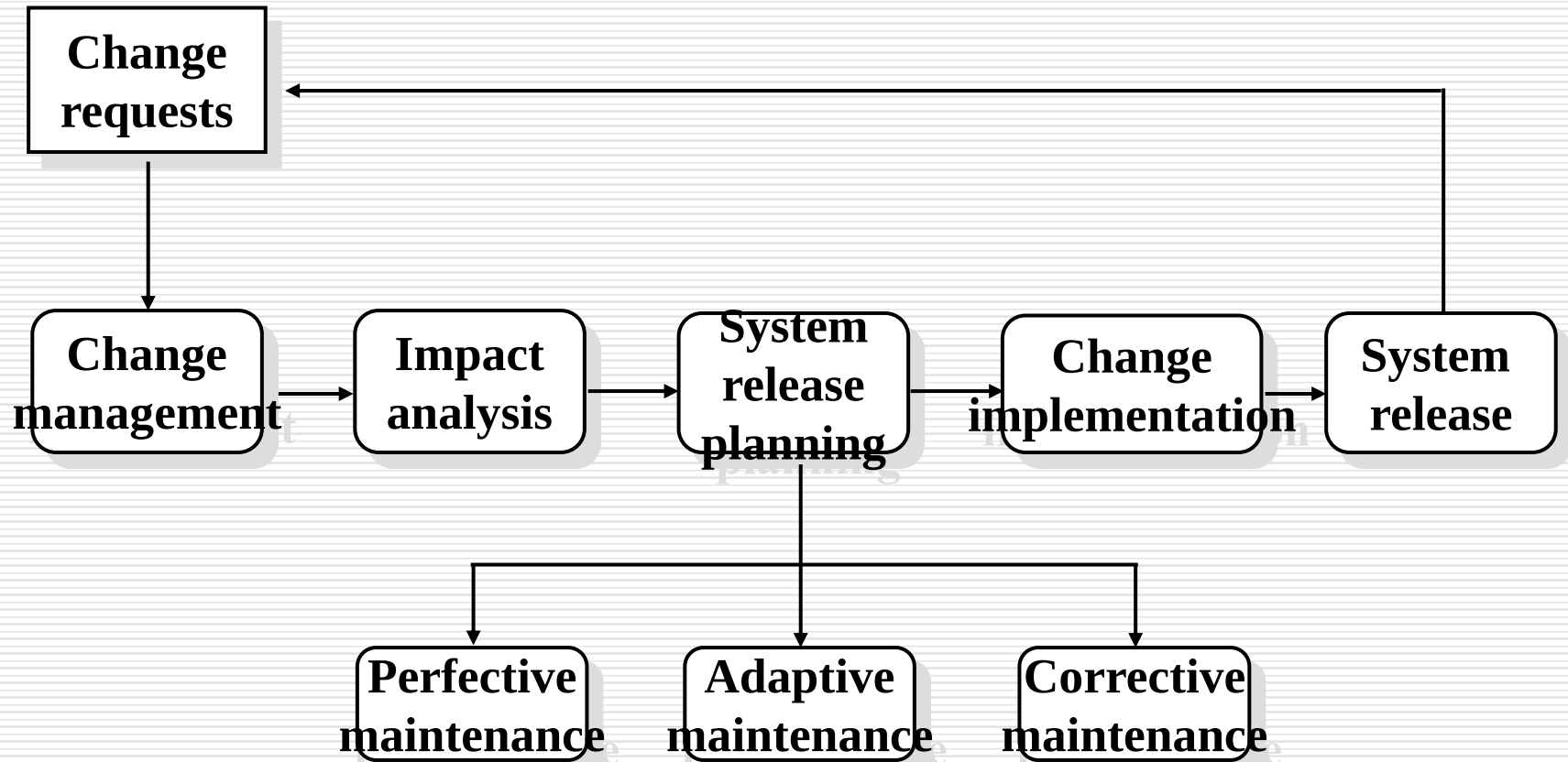
- ❑ “抓着谁就是谁”不可取
- ❑ 好的组织模式极为重要



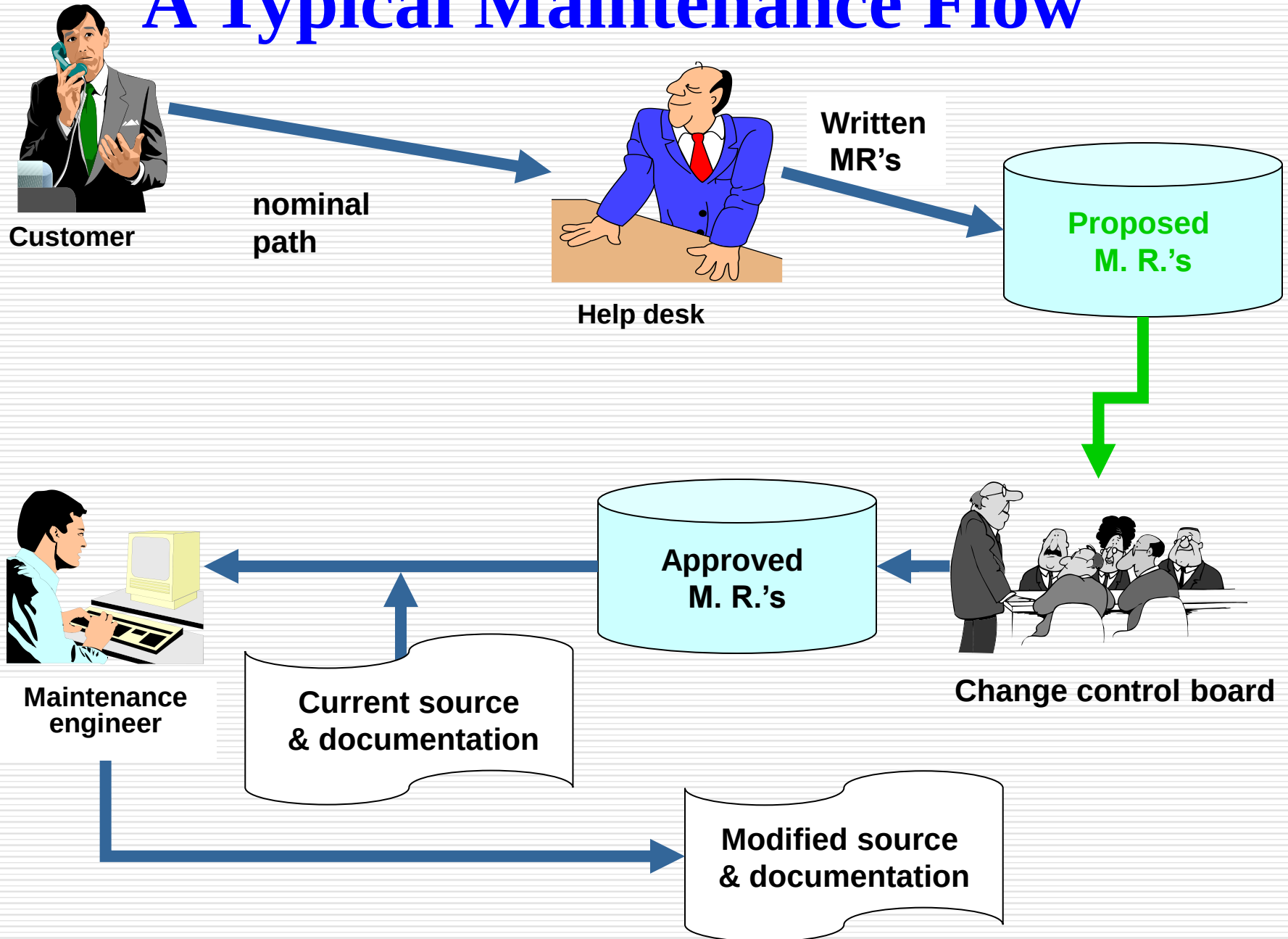
维护申请

- ✓ 维护申请报告是由软件组织外部提交的文档，它是计划维护活动的基础。软件组织内部应依此制定相应的软件修改报告，这个报告包括以下内容：
 - (1) 为满足某个维护申请要求所需的工作量；
 - (2) 所需修改变动的性质；
 - (3) 申请修改的优先级；
 - (4) 与修改有关的事后数据。
- ✓ 所有维护申请都应以标准化的文档形式
- ✓ 软件修改报告应提交修改负责人进行审核批准，以便进行下一步工作。

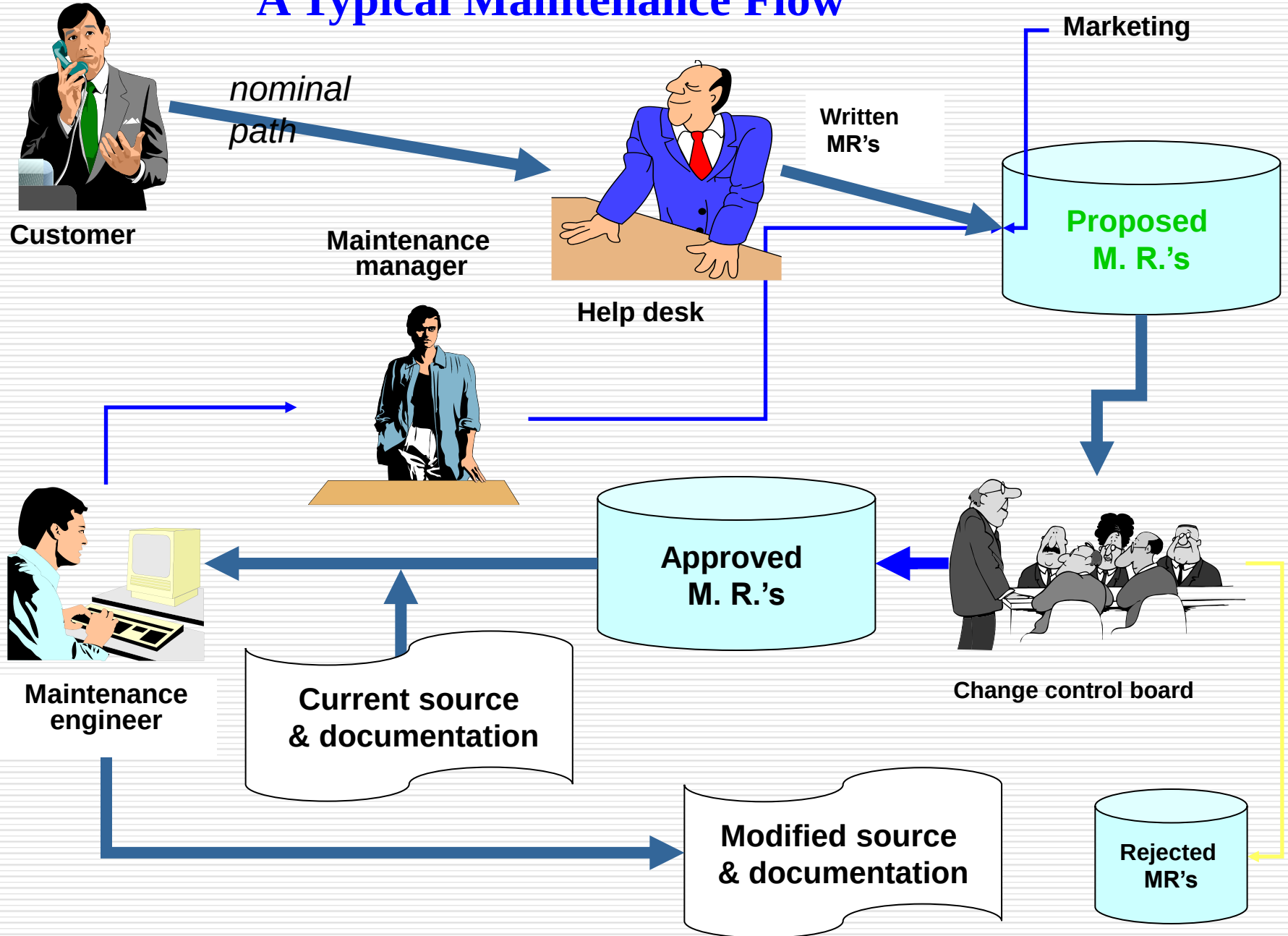
维护流程



A Typical Maintenance Flow



A Typical Maintenance Flow



The Maintenance

□ Change Management

- ✓ Uniquely identify, describe and track the status of all change requests

□ Impact Analysis

- ✓ identifies all systems and system products affected by a change request
- ✓ makes an estimate of the resources needed to effect the change
- ✓ analyses the benefits of the change

□ System Release Planning

- ✓ to establish the schedule and contents of a system release
- ✓ don't want each change request released as they are processed

The Maintenance

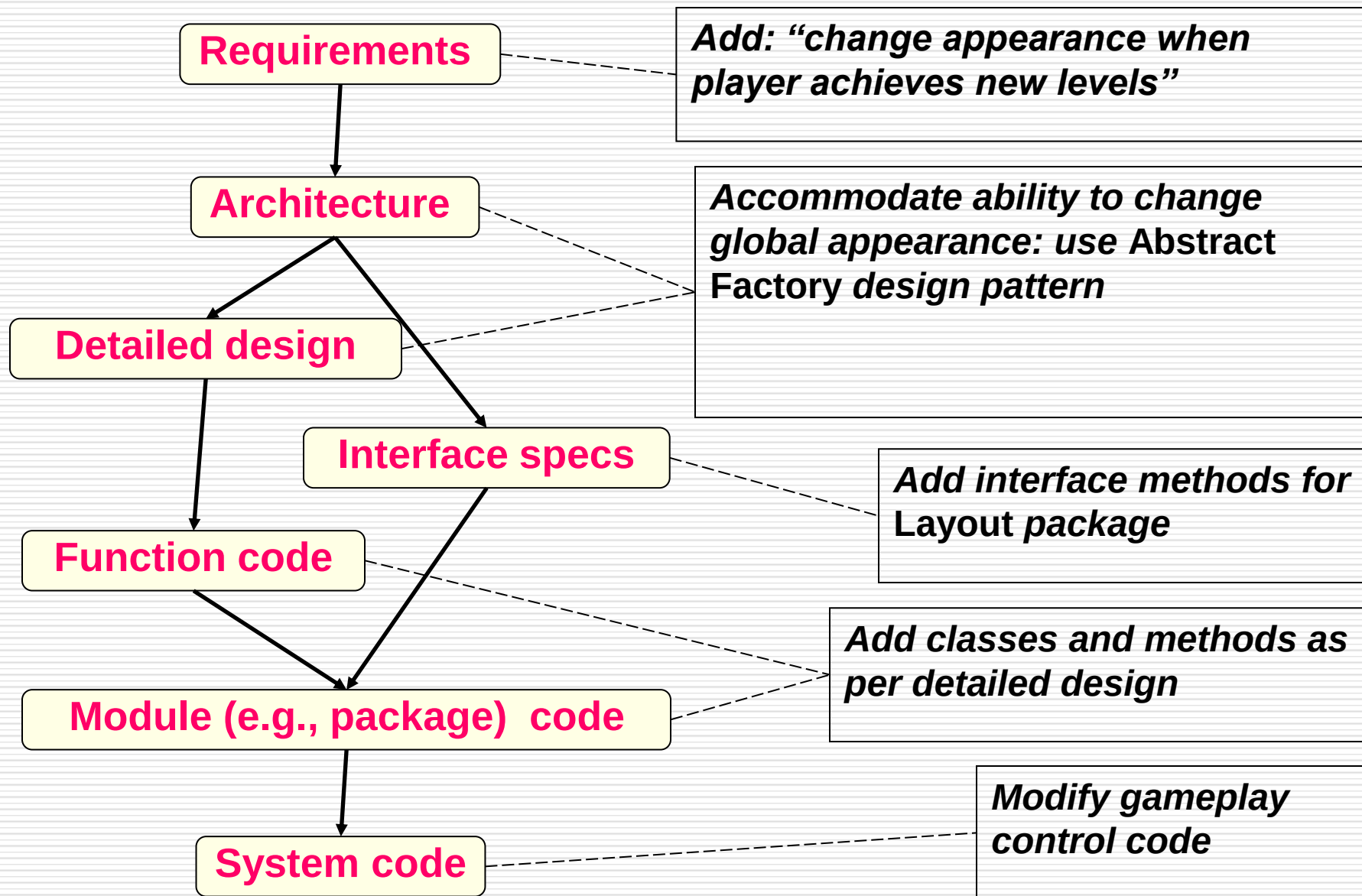
☐ Change Implementation

- ✓ Design Changes
- ✓ Coding
- ✓ Testing - must perform regression testing

☐ System Release includes

- ✓ documentation
- ✓ software
- ✓ training
- ✓ hardware changes
- ✓ data conversion

小型化的软件开发过程



维护记录

- ✓ 对什么进行了维护 (what)
- ✓ 做了什么修改和调整 (how)
- ✓ 谁及什么时候做的修改 (who & when)
- ✓ 维护所耗费的成本

维护记录

对于维护记录中的内容，Swanson给出了下述的项目表：

- (1) 程序名称；
- (2) 源程序语句条数；
- (3) 机器代码指令条数；
- (4) 使用的程序设计语言；
- (5) 程序的安装日期；
- (6) 程序安装后的运行次数；
- (7) 与程序安装后运行次数有关的处理故障的次数；

- (8) 程序修改的层次和名称;
- (9) 由于程序修改而增加的源程序语句条数;
- (10) 由于程序修改而删除的源程序语句条数;
- (11) 每项修改所付出的“人时”数;
- (12) 程序修改的日期;
- (13) 软件维护人员的姓名;
- (14) 维护申请报告的名称;
- (15) 维护类型;
- (16) 维护开始时间和维护结束时间;
- (17) 用于维护的累计“人时”数;
- (18) 维护工作的净收益。

维护评估

一般来说，可以从以下七个方面来评价维护工作：

- (1) 每次程序运行时的平均出错次数；
- (2) 用于每一类维护活动的总“人时”数；
- (3) 每个程序、每种维护类型所做的平均修改数；
- (4) 维护过程中，增加或删除每条源程序语句花费的平均“人时”数；
- (5) 用于每种语言的平均“人时”数；
- (6) 一张维护申请报告的平均处理时间；
- (7) 各类维护类型所占的百分比。

再工程 (Re-Engineering)

